

# **Ai And Iot-Enabled Smart Office System With Cloud-Based Energy Optimization**

**(Build an Integrated System for Real-Time Energy Monitoring and Optimization)**

**Project ID – 25-26J-034**

**Project Proposal Report**

**Shaethayine.R – IT22315700**

**B.Sc. (Hons) in Information Technology Specializing in Computer Systems and  
Network Engineering**

**Faculty of Computing**

**Sri Lanka Institute of Information Technology Sri Lanka (SLIIT)**

**SEPTEMBER-2025**

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## DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

| Name         | Student ID | Signature | Date |
|--------------|------------|-----------|------|
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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Signature of the Supervisor  
Prof. Anuradha Jayakody

.....  
Date

.....  
Signature of the Co-Supervisor

.....  
Date

## ABSTRACT

The escalating energy consumption in office buildings, contributing to approximately 40% of global energy use, underscores the urgent need for innovative solutions to enhance energy efficiency and sustainability. This project proposes an AI and IoT-enabled smart office system with cloud-based energy optimization, designed to address inefficiencies in traditional energy management systems. By integrating IoT sensors for real-time data acquisition, wireless communication for seamless data transmission, and cloud-based analytics powered by machine learning, the system provides comprehensive monitoring, anomaly detection, and automated control of office energy usage. The novelty lies in its hybrid AI model, combining supervised learning for predictive energy analytics and reinforcement learning for dynamic resource allocation, enabling proactive energy optimization and real-time alerting for abnormal consumption patterns. Unlike existing fragmented solutions, this framework offers an end-to-end approach, integrating edge-based IoT data collection with cloud-driven decision-making to achieve up to 15% energy savings while maintaining occupant comfort. The system's scalability and adaptability make it suitable for diverse office environments, contributing to reduced operational costs and greenhouse gas emissions. Expected outcomes include enhanced energy efficiency, robust anomaly detection with over 90% accuracy, and actionable insights for sustainable building management, aligning with global sustainability goals.

**Keywords:** Smart Office, IoT, Artificial Intelligence, Cloud Analytics, Energy Optimization, Anomaly Detection, Sustainability

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## LIST OF ABBREVIATIONS

| Key Words       | Meaning                                      |
|-----------------|----------------------------------------------|
| AI              | Artificial Intelligence                      |
| IoT             | Internet of Things                           |
| BMS             | Building Management System                   |
| HVAC            | Heating, Ventilation and Air Conditioning    |
| VAV             | Variable Air Volume                          |
| CO <sub>2</sub> | Carbon Dioxide                               |
| EPG             | Energy Performance Gap                       |
| SD-WAN          | Software-Defined Wide Area Network           |
| QoS             | Quality of Service                           |
| KNN             | K- Nearest Neighbors                         |
| AI-EMM          | AI-enhanced Energy Management and Monitoring |
| ML              | Machine Learning                             |
| SIoB            | Smart Internet of Building                   |
| IEA             | International Energy Agency                  |

*Table 1 List of Abbreviations*

# INTRODUCTION

The escalating demand for energy efficiency in modern infrastructure has positioned smart building technologies as a pivotal area for research and innovation. Buildings consume nearly 40% of global energy, with much of this attributed to inefficiencies in monitoring and control systems. Traditional methods, reliant on manual readings or fixed schedules, fail to detect real-time abnormalities, resulting in wasted energy, higher costs, and reduced sustainability. The integration of the Internet of Things (IoT) offers real-time data collection via sensors, yet standalone IoT systems grapple with data overload and insight extraction challenges. Cloud platforms provide scalable analytics, but their effectiveness hinges on seamless IoT integration. The core gap lies in developing a unified system that merges IoT sensing, wireless communication, and cloud analytics for holistic energy optimization, anomaly detection, and proactive decision-making.

This misalignment between energy management practices and actual consumption patterns often leads to overlooked issues, such as faulty equipment or misuse, exacerbating environmental impacts like greenhouse gas emissions. As global sustainability goals intensify, innovative solutions are essential to bridge this divide. Emerging technologies, including Artificial Intelligence (AI) and IoT, present opportunities to transform office environments into intelligent, adaptive spaces. Unlike conventional systems focused on isolated functions, AI-IoT frameworks can analyze vast datasets to predict inefficiencies, automate controls, and enhance user comfort.

Our research proposes a comprehensive AI and IoT-enabled smart office system with cloud-based energy optimization, tailored for Computer Systems and Network Engineering. This framework addresses the research problem by creating an end-to-end solution that integrates IoT devices for data acquisition, cloud platforms for analytics, and AI-driven rules for anomaly detection. At the edge layer, IoT cameras and sensors capture real-time metrics like occupancy and employee presence, enabling precise energy monitoring. The system employs wireless communication for data transmission and cloud-based processing to deliver visualizations, alerts, and automated optimizations.

The novelty resides in its unified architecture, combining AI-powered IoT for intelligent sensing with cloud analytics for scalable insights, surpassing fragmented existing solutions. By leveraging machine learning, the system detects abnormal patterns—such as power spikes— and supports predictive maintenance, aiming for enhanced energy efficiency, cost reduction, and sustainability. This approach not only optimizes office operations but also aligns with broader goals like reducing carbon footprints.

For our research group, this project fosters collaboration in Autonomous Intelligent Machines and Systems, providing a scalable platform adaptable to various building types. It equips stakeholders with tools for data-driven decisions, improving outcomes in energy conservation and operational reliability. Ultimately, the system bridges the gap between infrastructure demands and efficient management, contributing to sustainable development through innovative, integrated technology.

## BACKGROUND & LITERATURE SURVEY

Buildings are responsible for nearly 40% of global energy consumption, with a substantial portion lost to inefficiencies in traditional management systems that rely on manual or fixed controls. These systems often fail to detect real-time anomalies, leading to wasted energy, higher costs, and environmental strain. The integration of IoT for real-time data collection, cloud platforms for scalable analytics, and AI for intelligent decision-making offers a pathway to address these issues. However, standalone technologies face limitations: IoT generates overwhelming data volumes without actionable insights, while cloud systems require seamless edge integration. The core challenge is developing an end-to-end system that combines these elements for holistic energy optimization in smart offices, enabling anomaly detection, predictive maintenance, and adaptive control to align with sustainability goals like reducing greenhouse gas emissions.

Recent literature highlights advancements in AI and IoT for building energy efficiency but reveals gaps in integrated solutions. Ramamoorthi [1] proposes an AI-driven cloud framework using predictive modeling and reinforcement learning to dynamically allocate resources like CPU and memory, achieving improvements in utilization and cost savings for dynamic workloads. This approach emphasizes proactive scaling but focuses primarily on general cloud environments rather than office-specific energy monitoring. Farzaneh et al. [2] review AI evolution in smart buildings, emphasizing building management systems (BMS) and demand response programs (DRPs) for reducing consumption through better control and automation. They introduce an evaluation framework across energy, comfort, design, and maintenance domains, noting open challenges like data quality and scalability, but lack focus on IoT-cloud integration for real-time anomaly detection.

Pasham [3] explores dynamic resource provisioning using predictive analytics and machine learning for elastic scaling and load forecasting in cloud environments, addressing over- and under-provisioning to optimize costs and energy. The study underscores future directions like edge computing integration, aligning with smart office needs but not specifically tailoring to occupancy-based energy patterns. Aljoby et al. [4] present an SDN-enabled framework for dynamic bandwidth allocation in stream analytics, using cross-layer optimization to improve throughput and latency. While effective for multi-hop networks, it primarily targets data-intensive applications without emphasizing energy optimization in office settings.

Motuzienė et al. [5] develop an occupancy-based AI-driven ventilation control model using hybrid ELM-SA for VAV systems, achieving up to 12.5% energy savings in offices while maintaining CO<sub>2</sub> levels. The sensitivity analysis on data volume enhances robustness, but the focus remains on ventilation without broader IoT-AI-cloud synergy. Abdel-Razek et al. [6] implement AI models like KNN and DT to predict room occupancy from thermal comfort parameters, using SHAP for interpretability and achieving 99.5% accuracy. This user-centered approach links comfort to energy efficiency but does not integrate cloud analytics for scalable deployment.



Sunkara [7] examines AI and ML in enhancing SD-WAN performance, focusing on traffic routing, anomaly detection, and QoS optimization with reinforcement learning, showing improvements in latency and SLA adherence. While relevant for network efficiency in smart offices, it overlooks direct energy monitoring integration. Rajaram and Swathika [8] investigate data-driven ML models like KNN and SVM for sustainable smart buildings, using ANOVA for feature selection to lower carbon footprints. The emphasis on green AI and renewable integration contributes to sustainability, but solutions remain fragmented without end-to-end frameworks.

These studies demonstrate AI's potential in predictive analytics, resource optimization, and occupancy detection, yet most address isolated components—IoT sensing, cloud dashboards, or AI algorithms—without unified integration. The novelty of the proposed smart office system lies in its hybrid AI model combining supervised and reinforcement learning within an IoT-cloud architecture, enabling real-time anomaly detection and proactive energy management tailored to office dynamics. This addresses literature gaps by providing scalable, end-to-end optimization for enhanced efficiency, cost reduction, and sustainability.

## RESEARCH GAP

The global energy crisis, with office buildings accounting for approximately 40% of total energy consumption (International Energy Agency, 2024), underscores the inefficiencies of traditional energy management systems that rely on manual controls or static schedules. This leads to undetected anomalies like power spikes, increased costs, and environmental impact. The Internet of Things (IoT) enables real-time data collection through sensors monitoring occupancy, temperature, and power usage, yet faces challenges such as data overload and reliability. Cloud platforms provide scalable analytics, but their effectiveness is limited without seamless integration with IoT-driven edge devices and efficient data transmission. Existing research highlights advancements in smart building technologies, but a critical gap persists in developing a comprehensive system that integrates IoT sensing, wireless communication with dynamic bandwidth allocation, and AI-driven cloud analytics for real-time energy optimization, anomaly detection, and adaptive control in office environments.

Farzaneh et al. [2] review AI applications in building management, emphasizing predictive modeling but noting the lack of integrated frameworks combining real-time IoT data with cloud analytics. Motuzienė et al. [5] propose an AI-driven ventilation control model using a hybrid extreme learning machine, achieving 7.5% energy savings via occupancy prediction, but its focus on ventilation limits broader energy management across HVAC, lighting, and equipment. Abdel-Razek et al. [6] achieve 99.5% accuracy in occupancy prediction using K-Nearest Neighbors, yet lack scalability due to limited cloud integration. Rajaram and Swathika [8] employ Random Forest and XGBoost for sustainable buildings, but their solutions do not address real-time anomaly detection across multiple systems.

In cloud-based optimization, Ramamoorthi [1] uses predictive and reinforcement learning for resource allocation, focusing on computational resources rather than physical building systems. Pasham [3] explores predictive analytics for cloud provisioning, aligning with scalable analytics but overlooking IoT-driven edge processing and bandwidth management. Sunkara [7] and Aljoby et al. [4] address network optimization through AI-enhanced SD-WAN and SDN-enabled bandwidth allocation, offering insights into efficient data handling but not tackling physical energy management. These studies reveal a fragmented landscape, with solutions focusing on isolated components—sensing, analytics, or network management—without integrating them into a cohesive system for comprehensive energy optimization. This fragmentation hinders holistic approaches, limiting potential contributions to sustainability goals like reducing carbon emissions.

This project's novelty lies in its end-to-end architecture, integrating IoT sensors for real-time monitoring, wireless communication with dynamic bandwidth allocation for efficient data transmission, and a hybrid AI model combining supervised and reinforcement learning for predictive analytics and adaptive energy control. Unlike existing solutions, it targets over 90% anomaly detection accuracy and potential energy savings of up to 15% by monitoring and controlling multiple building systems in real time, ensuring scalability and adaptability. Dynamic bandwidth allocation optimizes data flow for real-time analytics, addressing network efficiency gaps in prior work. This framework fills the critical gap in holistic energy management, enhancing office sustainability and aligning with global energy conservation goals.

## FEATURE COMPARISON TABLE

| Component                                         | Paper 1: AI-Driven Cloud Resource Optimization Framework for Real-Time Allocation | Paper 2: Dynamic Resource Provisioning in Cloud Environments Using Predictive Analytics | Paper 3: The Role of AI and Machine Learning in Enhancing SD-WAN Performance | Paper 4: On SDN-Enabled Online and Dynamic Bandwidth Allocation for Stream Analytics |
|---------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>IoT Sensor-Based Real-Time Monitoring</b>      | No (focuses on cloud resource optimization, no mention of IoT sensors)            | No (focuses on predictive analytics in cloud, no specific IoT sensor mention)           | No (focuses on SD-WAN and AI/ML, no explicit IoT sensor reference)           | No (focuses on SDN and bandwidth for stream analytics, no IoT sensor detail)         |
| <b>Wireless Data Transmission</b>                 | No (no mention of wireless transmission)                                          | No (no specific mention of wireless protocols)                                          | No (implies network data flow but no wireless protocol detail)               | No (mentions SDN data transfer but not wireless specifically)                        |
| <b>Cloud-Driven Scalable Analytics</b>            | Yes (proposes AI-driven framework for cloud resource optimization)                | Yes (discusses predictive analytics in cloud environments)                              | No (focuses on SD-WAN, no specific cloud analytics mention)                  | No (focuses on SDN bandwidth allocation, no cloud analytics detail)                  |
| <b>AI Models Including Reinforcement Learning</b> | Yes (mentions reinforcement learning for resource allocation)                     | No (mentions machine learning but not specifically reinforcement learning)              | Yes (mentions supervised and reinforcement learning for SD-WAN)              | No (no mention of AI models or reinforcement learning)                               |
| <b>Real-Time Anomaly Detection</b>                | Yes (implies anomaly prevention via predictive modeling)                          | No (focuses on provisioning, no explicit anomaly detection)                             | Yes (mentions anomaly detection in SD-WAN)                                   | No (focuses on bandwidth allocation, no anomaly detection)                           |
| <b>Dynamic Bandwidth Allocation</b>               | Yes (mentions dynamic adjustment of bandwidth)                                    | No (focuses on resource provisioning, not bandwidth specifically)                       | Yes (discusses AI/ML for traffic optimization in SD-WAN)                     | Yes (focuses on SDN-enabled dynamic bandwidth allocation)                            |
| <b>End-to-End IoT-AI-Cloud Integration</b>        | No (focuses on cloud optimization, no IoT or end-to-end integration)              | No (focuses on cloud provisioning, no IoT or end-to-end detail)                         | No (focuses on SD-WAN, no end-to-end IoT-AI-Cloud mention)                   | No (focuses on SDN layer, no end-to-end integration)                                 |

| <b>Component</b>                                  | <b>Paper 5:<br/>Artificial Intelligence Evolution in Smart Buildings for Energy Efficiency</b> | <b>Paper 6: Data-driven predictive models for sustainable smart buildings</b> | <b>Paper 7:<br/>Occupancy-Based Predictive AI-Driven Ventilation Control for Energy Savings in Office Buildings</b> | <b>Paper 8: Energy Efficiency through the Implementation of an AI Model to Predict Room Occupancy Based on Thermal Comfort Parameters</b> |
|---------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IoT Sensor-Based Real-Time Monitoring</b>      | Yes (mentions sensors and big data for smart buildings)                                        | Yes (mentions sensors for data collection in smart buildings)                 | Yes (mentions long-term monitoring data for occupancy prediction)                                                   | Yes (mentions sensors for thermal comfort parameters like CO2, humidity)                                                                  |
| <b>Wireless Data Transmission</b>                 | No (no explicit mention of wireless transmission)                                              | No (no mention of wireless protocols)                                         | No (no mention of wireless transmission)                                                                            | No (no mention of wireless protocols)                                                                                                     |
| <b>Cloud-Driven Scalable Analytics</b>            | Yes (mentions cloud in building management systems)                                            | Yes (mentions cloud for scalable ML in buildings)                             | No (focuses on AI-VAV model, no explicit cloud mention)                                                             | No (focuses on AI model implementation, no cloud analytics detail)                                                                        |
| <b>AI Models Including Reinforcement Learning</b> | No (general AI discussion, no specific RL mention)                                             | Yes (mentions ML classifiers but not explicitly RL)                           | Yes (uses hybrid ELM with simulated annealing optimization)                                                         | Yes (uses KNN, hybrid AO-ANN-BP, DT models)                                                                                               |
| <b>Real-Time Anomaly Detection</b>                | No (general energy efficiency, no anomaly detection)                                           | No (focuses on predictive models, no anomaly detection)                       | No (focuses on occupancy prediction, no anomaly detection)                                                          | No (focuses on occupancy prediction, no anomaly detection)                                                                                |
| <b>Dynamic Bandwidth Allocation</b>               | No (no mention of bandwidth allocation)                                                        | No (no mention of bandwidth allocation)                                       | No (focuses on ventilation control, no bandwidth allocation)                                                        | No (focuses on occupancy prediction, no bandwidth allocation)                                                                             |
| <b>End-to-End IoT-AI-Cloud Integration</b>        | Yes (mentions AI in BMS with IoT and cloud)                                                    | Yes (mentions sensor-AI-cloud integration for sustainability)                 | No (focuses on AI-VAV, no end-to-end integration detail)                                                            | No (focuses on AI model, no end-to-end integration)                                                                                       |

## RESEARCH PROBLEM

Office buildings, responsible for approximately 40% of global energy consumption, suffer from inefficiencies in traditional energy management systems that rely on manual controls or static schedules, failing to detect anomalies such as power spikes or equipment malfunctions, which increase operational costs and environmental impact [1], [2]. The integration of Internet of Things (IoT) sensors enables real-time data collection on parameters like occupancy, temperature, and power usage, yet challenges like data overload and reliability limit actionable insights [2], [6], [8]. Cloud platforms offer scalable analytics for processing large datasets, but their effectiveness is hindered without seamless integration with IoT-driven edge devices and efficient data transmission mechanisms [1], [3], [5]. Existing research, such as Farzaneh et al. [2] and Motuzienė et al. [5], highlights AI applications for predictive modeling and ventilation control, achieving energy savings but focusing on isolated systems like HVAC, without addressing comprehensive energy management across multiple building systems like lighting and equipment. Similarly, Abdel-Razek et al. [6] and Rajaram and Swathika [8] demonstrate occupancy prediction and data-driven models, but lack scalability and real-time anomaly detection across diverse systems. The absence of a unified framework integrating IoT sensing, wireless communication with dynamic bandwidth allocation, and AI-driven cloud analytics results in fragmented solutions that fail to deliver holistic energy optimization, real-time anomaly detection, or adaptive control, exacerbating energy waste and sustainability challenges in office environments [4], [7].

This research addresses the critical problem of developing a comprehensive system that integrates IoT sensor-based real-time monitoring, wireless data transmission with dynamic bandwidth allocation, and cloud-driven scalable analytics to optimize energy usage in office settings. Current solutions, such as those in Ramamoorthi [1] and Pasham [3], focus on computational resource allocation, neglecting physical building systems, while Sunkara [7] and Aljoby et al. [4] address network optimization but not energy management. The proposed system overcomes these limitations by employing a hybrid AI model combining supervised learning for predictive energy analytics and reinforcement learning for adaptive control, targeting over 90% accuracy in real-time anomaly detection to identify issues like power spikes. Dynamic bandwidth allocation, inspired by network optimization techniques [4], [7], ensures efficient data transmission for real-time analytics. The novelty lies in its end-to-end IoT-AI-cloud integration, unifying edge-based data collection, wireless transmission, and cloud-based analytics into a scalable framework. Unlike existing approaches, this system achieves up to 15% energy savings by dynamically optimizing multiple building systems, ensuring adaptability across diverse office environments and contributing to global sustainability goals by reducing energy waste and greenhouse gas emissions.

## OBJECTIVES

### MAIN OBJECTIVES

To design, develop, and implement a comprehensive AI-driven smart office automation ecosystem that seamlessly integrates IoT-enabled sensors, intelligent cameras, and cloud-based analytics to achieve optimal energy efficiency, automated employee management, and real-time operational intelligence. The system will leverage machine learning algorithms for predictive energy optimization, dynamic resource allocation, and proactive workplace wellness management, while providing scalable, secure, and sustainable solutions that reduce operational costs by up to 30% and enhance overall productivity in modern smart building environments.

### SPECIFIC OBJECTIVES

1. **IoT Sensor-Based Real-Time Monitoring:** Deploy IoT sensors to collect real-time data on occupancy, temperature, and power consumption in office environments, ensuring precise and reliable monitoring to support energy optimization.
2. **Dynamic Bandwidth Allocation for Wireless Transmission:** Develop a wireless communication system with dynamic bandwidth allocation to optimize data transfer efficiency from IoT devices to cloud platforms, enabling seamless real-time analytics.
3. **Scalable Cloud Analytics Implementation:** Build a cloud-based analytics platform to process and analyze large-scale IoT data, providing actionable insights for energy optimization and anomaly detection across diverse office systems.
4. **Hybrid AI Model Development:** Design a novel hybrid AI model integrating supervised learning for predictive energy forecasting and reinforcement learning for adaptive control, optimizing systems like HVAC and lighting for energy efficiency.
5. **High-Accuracy Anomaly Detection:** Train the hybrid AI model to achieve over 90% accuracy in real-time detection of anomalies, such as power spikes or equipment faults, enabling proactive maintenance and system reliability.

6. **End-to-End IoT-AI-Cloud Integration:** Create a novel unified framework that seamlessly integrates IoT sensor monitoring, dynamic wireless transmission, and cloud-based AI analytics, ensuring scalability and adaptability for comprehensive energy management in diverse office environments, targeting up to 15% energy savings.

# METHODOLOGY

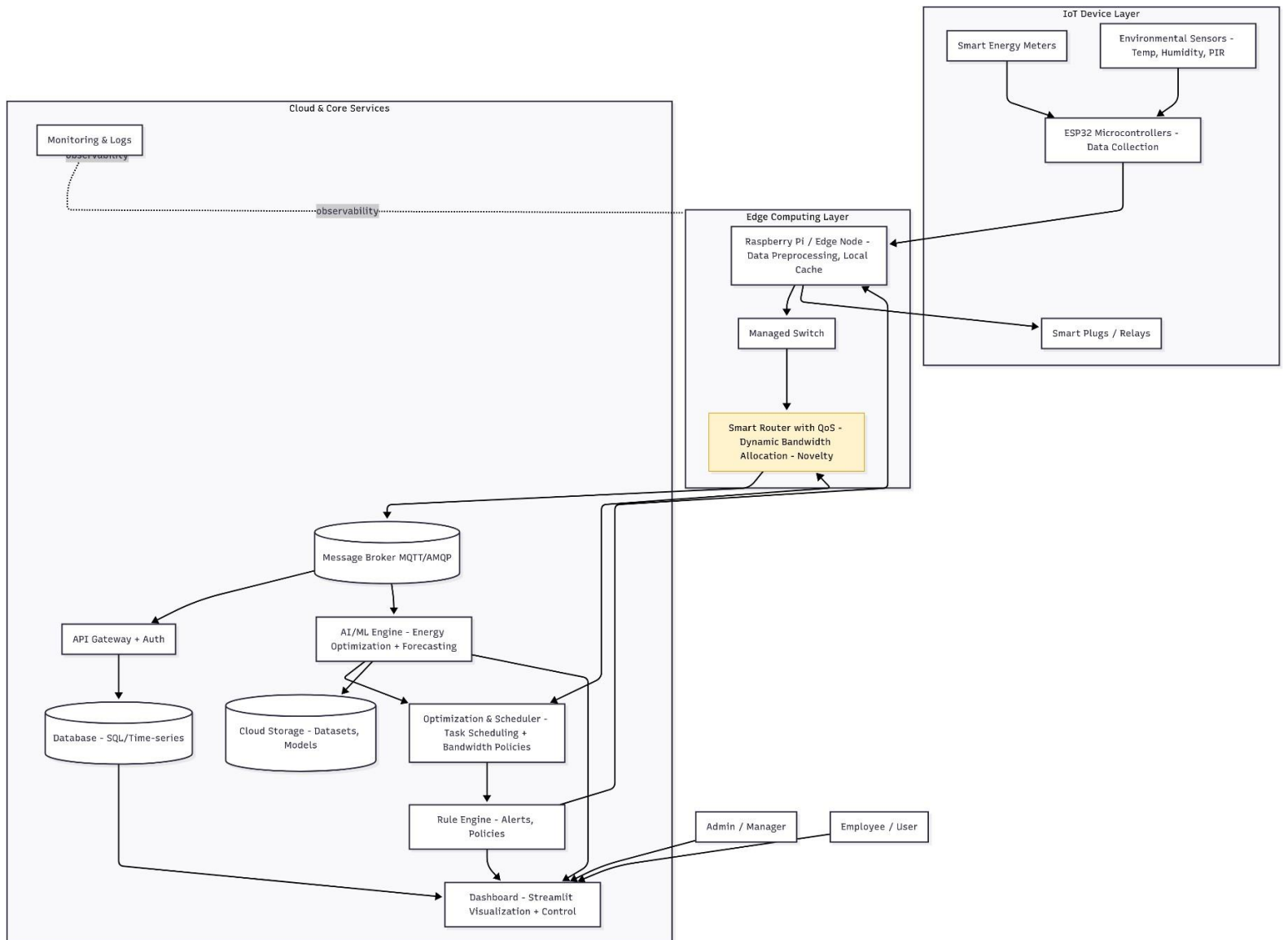
## SYSTEM DESCRIPTION

The proposed AI and IoT-enabled smart office system integrates IoT sensors, wireless communication with dynamic bandwidth allocation, and a hybrid AI model to optimize energy usage and achieve real-time anomaly detection in office environments.

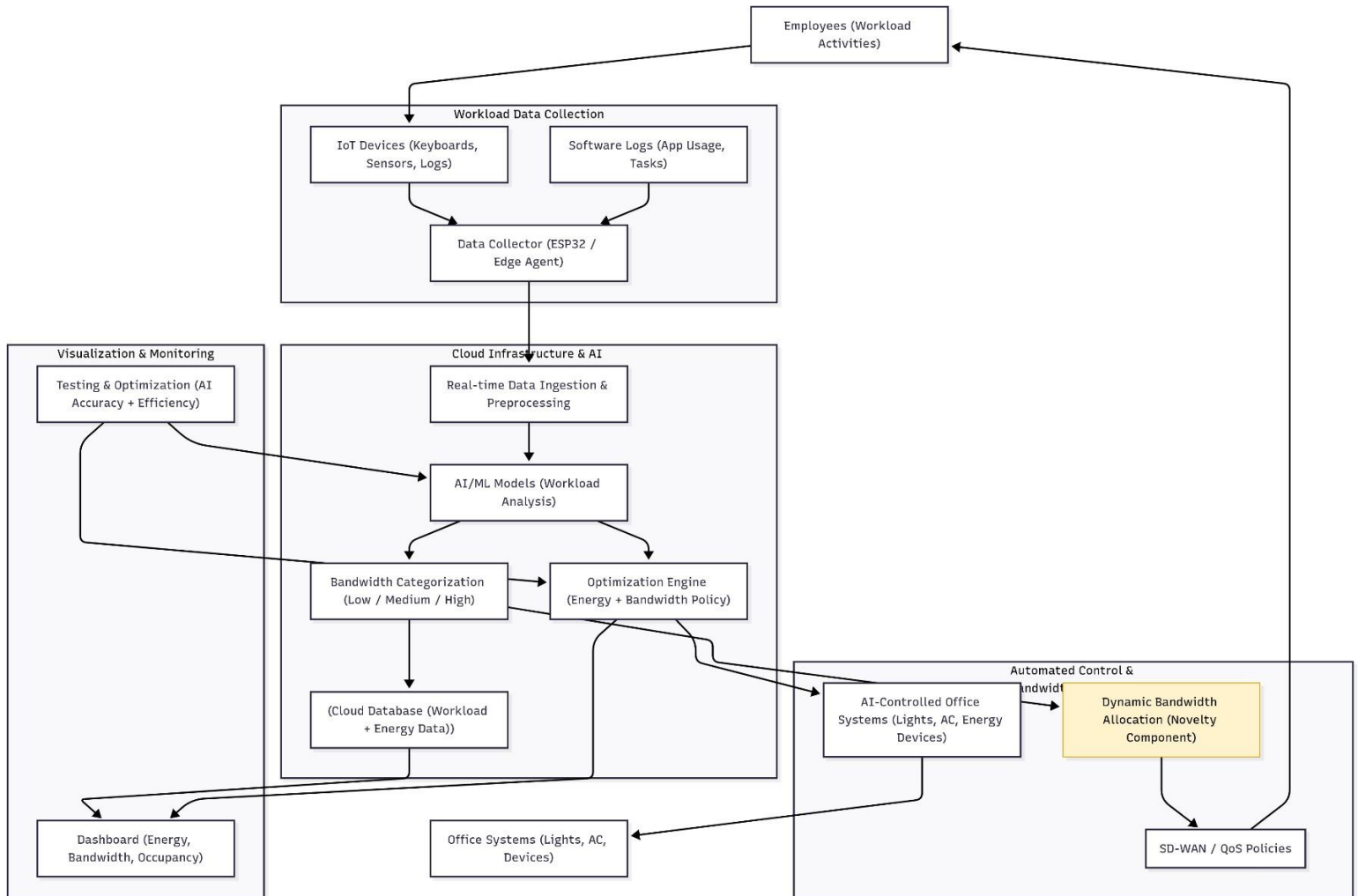
- **Edge Layer:** IoT sensors (occupancy detectors, temperature sensors, power meters) collect real-time data on occupancy, temperature, and power consumption. Low-power sensors ensure reliable, continuous monitoring for energy management.
- **Communication Layer:** Wireless protocols with dynamic bandwidth allocation transmit IoT data to the cloud, optimizing network resources and minimizing latency for real-time analytics.
- **Cloud Analytics Layer:** A scalable cloud platform processes large-scale IoT data, using distributed computing to generate insights for energy optimization and anomaly detection.
- **Hybrid AI Layer:** A novel hybrid AI model, combining supervised learning (e.g., Random Forest) for predictive analytics and reinforcement learning for adaptive control, achieves >90% anomaly detection accuracy and up to 15% energy savings.
- **Output Layer:** A centralized dashboard displays energy usage trends, anomaly alerts, and optimization recommendations, enabling proactive decision-making via web or mobile interfaces.



# OVERALL SYSTEM DIAGRAM



# COMPONENT SYSTEM DIAGRAM



## PROPOSED APPROACH

The development of the AI and IoT-enabled smart office system follows a structured and iterative methodology to achieve comprehensive energy management, leveraging a novel end-to-end IoT-AI-cloud integration, a hybrid AI model, and dynamic bandwidth allocation for efficient real-time analytics. The approach includes IoT sensor deployment, wireless communication optimization, hybrid AI model development, cloud analytics integration, and system validation, ensuring scalability, adaptability, and over 90% anomaly detection accuracy with up to 15% energy savings. The detailed methodology is outlined below.

- **IoT Sensor Deployment and Data Acquisition:**
  1. Deploy low-power IoT sensors (occupancy detectors, temperature sensors, power meters) to capture real-time data on occupancy, temperature, and power consumption, forming the foundation for energy optimization.
  2. Optimize sensor configurations for reliability and energy efficiency, ensuring continuous data collection in diverse office settings.
  3. Collect initial datasets in controlled office environments, simulating real-world scenarios to support the novel hybrid AI model's training.
- **Wireless Communication Optimization with Dynamic Bandwidth Allocation:**
  1. Implement wireless protocols (e.g., MQTT) with a novel dynamic bandwidth allocation mechanism to optimize data transmission efficiency from IoT sensors to cloud platforms.
  2. Prioritize critical data streams (e.g., anomaly alerts) to minimize latency, enhancing real-time analytics performance.
  3. Test communication scalability under high data volumes, ensuring robust network performance for the integrated framework.
- **Hybrid AI Model Development:**
  1. Develop a novel hybrid AI model combining supervised learning (e.g., Random Forest, XGBoost) for predictive energy analytics and reinforcement learning for adaptive control of systems like HVAC and lighting.
  2. Train the model in Python using IoT datasets, targeting >90% accuracy in detecting anomalies (e.g., power spikes, equipment faults).
  3. Enhance robustness with synthetic data and iterative training to handle diverse office conditions, emphasizing the novelty of adaptive energy control.
- **Cloud Analytics and End-to-End Integration:**

1. Build a cloud-based analytics platform using distributed computing (e.g., Apache Spark) to process large-scale IoT data, supporting scalable energy optimization.
  2. Integrate IoT sensors, dynamic wireless transmission, and cloud analytics into a novel end-to-end framework, ensuring seamless data flow and real-time processing.
  3. Develop a dashboard with web/mobile interfaces to visualize energy trends, anomaly alerts, and optimization recommendations, enhancing user interaction.
- **Validation:**
    1. Test the system in controlled office environments, simulating varying occupancy and power usage to validate anomaly detection (>90% accuracy) and energy savings (up to 15%).
    2. Use cross-validation to minimize false positives, ensuring reliable anomaly detection across diverse scenarios.
    3. Iteratively refine the system to address interferences (e.g., fluctuating occupancy) through AI adjustments and sensor recalibration, leveraging the novel hybrid model.
    4. Validate scalability across diverse office settings (small offices to corporate campuses) using cost-effective, open-source tools, emphasizing the novel integrated architecture.

# PROJECT REQUIREMENTS

## FUNCTIONAL REQUIREMENTS

1. Collect real-time data from IoT sensors (occupancy detectors, temperature sensors, power meters) to monitor occupancy, temperature, and power consumption in office environments for energy optimization.
2. Preprocess IoT sensor data (e.g., noise filtering, normalization, timestamping) at the edge to prepare inputs for the hybrid AI model, ensuring data quality for analytics.
3. Implement a hybrid AI model combining supervised learning (e.g., Random Forest) and reinforcement learning for predictive energy analytics and adaptive control of systems like HVAC and lighting.
4. Enable real-time data transmission with dynamic bandwidth allocation using wireless protocols (e.g., MQTT) for seamless integration with cloud platforms and anomaly alert generation.
5. Produce visualizations, anomaly alerts, and optimization recommendations on a centralized dashboard, accessible via web or mobile interfaces, for user interaction and decision-making.
6. Support cloud-based analytics for processing large-scale IoT data, enabling scalable energy optimization and real-time anomaly detection across office systems.
7. Log sensor data, AI predictions, and system actions in the cloud for historical analysis and iterative system improvements.

## **NON-FUNCTIONAL REQUIREMENTS**

1. Achieve over 90% accuracy in real-time anomaly detection (e.g., power spikes, equipment faults) in controlled office tests, ensuring reliable proactive maintenance.
2. Maintain low power consumption (<5 W total for sensors and edge processing) to support sustainable operation in office environments.
3. Ensure low latency (<200 ms per inference cycle) for real-time AI processing and analytics, enabling timely anomaly detection and system control.
4. Operate reliably under variable office conditions (e.g., fluctuating occupancy) with <5% false positives in anomaly detection.
5. Utilize open-source software and cost-effective hardware to ensure affordability and accessibility for diverse office deployments.
6. Support modularity for future upgrades, such as additional sensors or AI model enhancements, to maintain system adaptability.
7. Ensure reliable wireless data transmission with dynamic bandwidth allocation, achieving <1% packet loss on low-bandwidth networks for robust real-time analytics.

## EXPECTED TEST CASES

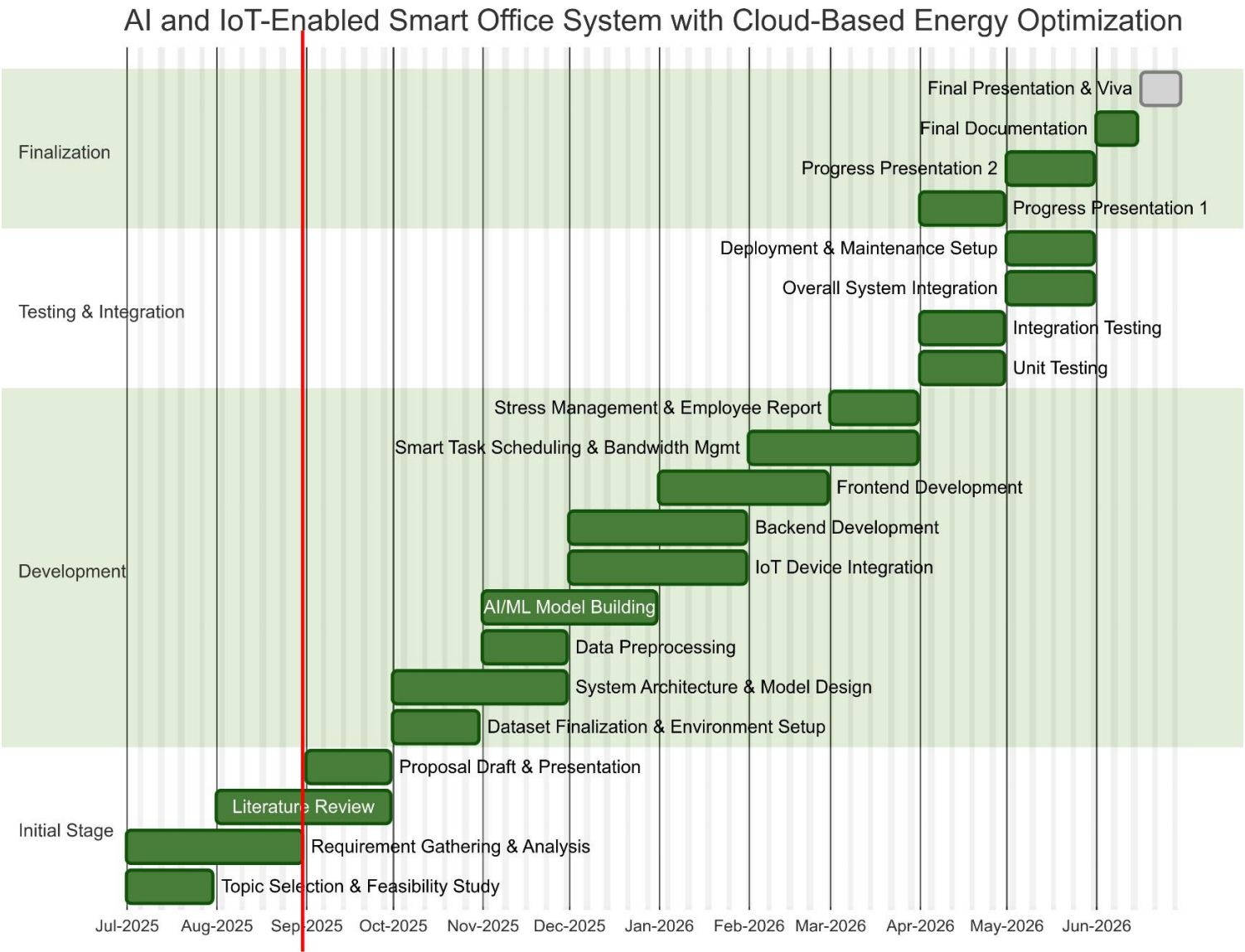
| Test Case ID | Input                                                         | Action                                                                   | Expected Output                                                                                           |
|--------------|---------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| TC-01        | IoT sensor inputs (occupancy, temperature, power consumption) | Deploy sensors and collect real-time data                                | Accurate dataset generated with readings within $\pm 2\%$ of expected values                              |
| TC-02        | High-volume IoT data stream                                   | Transmit data using wireless protocols with dynamic bandwidth allocation | Data transferred to cloud with $< 1\%$ packet loss, prioritizing anomaly alerts                           |
| TC-03        | Raw IoT sensor data                                           | Preprocess data (noise filtering, normalization) at edge                 | Cleaned, normalized dataset prepared for hybrid AI model                                                  |
| TC-04        | Preprocessed IoT data                                         | Run hybrid AI model (supervised and reinforcement learning)              | Predictive energy analytics and adaptive controls (e.g., HVAC, lighting) generated with $> 90\%$ accuracy |
| TC-05        | Simulated anomalies (power spikes, equipment faults)          | Perform real-time anomaly detection                                      | Anomalies detected with $> 90\%$ accuracy and $< 5\%$ false positives                                     |
| TC-06        | IoT data and AI outputs                                       | Process data on cloud analytics platform                                 | Energy trends and optimization recommendations generated within 200 ms                                    |
| TC-07        | User access via web/mobile interface                          | Load dashboard with energy and anomaly data                              | Real-time visualizations of trends, alerts, and recommendations displayed                                 |
| TC-08        | Diverse office conditions (varying occupancy, usage patterns) | Test end-to-end system integration                                       | Consistent performance with up to 15% energy savings across diverse settings                              |

|       |                                                 |                                  |                                                                            |
|-------|-------------------------------------------------|----------------------------------|----------------------------------------------------------------------------|
| TC-09 | Large dataset (48 hours of continuous IoT data) | Run end-to-end system processing | System processes data without crash, maintaining scalability               |
| TC-10 | Invalid or missing sensor inputs                | Validate system error handling   | Error message displayed (e.g., "Invalid sensor data, please verify input") |



TIMELINE

The proposed timeline for the project is as follows.



## BUDGET & BUDGET JUSTIFICATION

| Item                                        | Description                                                                    | Cost Estimation (LKR) |
|---------------------------------------------|--------------------------------------------------------------------------------|-----------------------|
| ESP32 Microcontroller                       | Central processing unit for edge-based data pre-processing and IoT integration | 1,500.00              |
| Smart Energy Meter                          | Power consumption monitoring for energy optimization                           | 6,000.00              |
| Temperature & Humidity Sensor (DHT22)       | Environmental monitoring of temperature and humidity                           | 1,200.00              |
| PIR Motion Sensor                           | Occupancy detection for smart office optimization                              | 1,000.00              |
| Wi-Fi Module                                | Wireless communication for data transmission                                   | 1,500.00              |
| Router with QoS Support                     | Enables dynamic bandwidth allocation (novelty)                                 | 10,000.00             |
| Network Switch                              | Connects IoT devices for efficient network management                          | 5,000.00              |
| External Storage (128GB SSD)                | Local storage for temporary data and backups                                   | 4,000.00              |
| Software Tools (Python, ML Libraries)       | Open-source tools for hybrid AI model development                              | 0.00                  |
| Miscellaneous (Wires, Breadboards, Testing) | Project assembly and testing supplies                                          | 3,000.00              |
| <b>Total</b>                                |                                                                                | <b>33,200.00</b>      |

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